

Haiyang Zhang

Senior Frontend / Full-Stack Engineer

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PROFESSIONAL SUMMARY

- Senior Frontend / Full-Stack Engineer with 8 years of experience building production web, mobile, AI-powered healthcare, backend API, and B2B workflow automation platforms.
- Early engineering hire at Lavita AI, contributing from startup-stage product foundations to scaled delivery across a 100,000+ download healthcare app and a \$5.9M enterprise workflow platform.
- Strong hands-on experience with JavaScript, TypeScript, React.js, React Native, Expo, Vue.js, Nuxt.js, Next.js, Node.js, Express.js, Python, FastAPI, REST APIs, GraphQL, PostgreSQL, Retrieval-Augmented Generation (RAG), and cloud deployment.
- Built production AI chat and retrieval workflows using hybrid semantic and keyword search, reranking, streaming responses, output validation, and bounded retry policies.
- Owned frontend delivery across 2 core enterprise subsystems: compute node management and data collaboration management, covering 123 scoped requirements and 107 implementation-complete items.
- Improved engineering efficiency with 30% to 40% less duplicated implementation, 30%+ workflow-canvas performance gains, 40% less manual regression testing, 35% fewer release-blocking regressions, and 70% smaller Electronic Health Record (EHR) payloads.

PROFESSIONAL EXPERIENCE

Lavita AI

Senior Frontend / Full-Stack Engineer and Early Engineering Hire | Sep 2018 - Present

Healthcare Mobile App Platform — AI Chat, RAG, and Health Data

Mobile, Backend API, and AI Product Delivery

- Owned core mobile product surfaces for a React Native and Expo healthcare app with 100,000+ downloads, shipping AI chat, health-data upload, onboarding, and account workflows across iOS and Android.
- Built Express.js middleware and Python/FastAPI REST APIs backed by PostgreSQL, supporting 15+ backend/API endpoints across AI chat, RAG retrieval, health-data processing, file processing, reporting, and account workflows.
- Built a Retrieval-Augmented Generation (RAG) pipeline for an AI health assistant, combining semantic search and keyword retrieval with reranking to surface relevant healthcare knowledge for LLM response generation.
- Designed Server-Sent Events (SSE) streaming for AI chat, reducing perceived response latency by 40% through incremental message rendering, loading-state management, and smoother real-time UI updates.
- Implemented AI output validation and bounded retry workflows, automatically re-running failed generations with feedback when responses did not meet defined quality criteria.
- Led Electronic Health Record (EHR) ingestion workflows across Epic, Cerner, and other systems, including Fast Healthcare Interoperability Resources (FHIR) transformation that reduced medical-record payload size by 70%.
- Expanded Jest coverage to 65%+ across 80+ utility, API, and UI-state functions and 12 critical health-data flows, reducing release-blocking regression frequency by 35% across 2 app-store release tracks.

Workflow Automation Platform — \$5.9M 8-Subsystem Enterprise Platform

Frontend Engineer and Workflow Platform Developer

- Designed and developed major frontend modules within a \$5.9M B2B workflow automation platform spanning 8 interconnected subsystems, owning compute node management and data collaboration management.

- Delivered 107 implementation-complete items across 123 scoped requirements, including 53 compute-node items and 54 data-collaboration items.
- Built Compute Node Management workflows across 61 scoped requirements, covering node connectivity, resource/container monitoring, structured and unstructured data sources, preprocessing, permissions, and logs.
- Built the complex workflow orchestration layer within Data Collaboration Management across 7 workflow modules and 30 completed workflow requirements, covering project/task creation, component-driven canvas editing, data-source and parameter configuration, execution tracking, model management, and result publishing.
- Expanded Cypress and Playwright coverage across 20+ critical workflow, node, data-source, and permission paths, reducing manual regression testing time by 40%.

Workflow Automation Platform Refactor — Standardized Customer Delivery Foundation

Senior Frontend and Platform Architecture Ownership

- Led the refactor of 2 subsystem families within the broader 8-subsystem workflow platform, converting patterns from 123 feature requirements into a reusable foundation for customer-specific delivery.
- Abstracted 10+ repeated product patterns into shared platform modules for Role-Based Access Control (RBAC), table/search/form flows, data-source registration, approval workflows, audit logs, monitoring dashboards, workflow tasks, and canvas interactions.
- Standardized 6 reusable workflow templates across project management, task management, component-driven workflow editing, data-source selection, execution tracking, and result/model publishing.
- Reduced duplicated implementation by 30% to 40% by consolidating shared components, data models, routing, permission checks, table/search patterns, and workflow interactions; conservatively estimated to save 0.8 to 1.5 frontend engineer-years and \$100K to \$180K in annual delivery effort.
- Raised frontend engineering standards for a 4-person frontend group within a 30-person delivery team through technical proposals, reusable component contracts, Webpack/Vite build patterns, onboarding, and code reviews.

TECHNICAL SKILLS

- Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3
- Frontend and Mobile: React.js, Next.js, React Native, Expo, Vue.js, Nuxt.js, Vue Router, D3.js, Tailwind CSS, Sass / SCSS
- Backend and APIs: Node.js, Express.js, FastAPI, REST APIs, RESTful APIs, GraphQL, PostgreSQL, Middleware Development, Database-Backed API Development
- AI and LLM Applications: Retrieval-Augmented Generation (RAG), Hybrid Retrieval, Semantic Search, Keyword Search, Reranking, LLM Context Assembly, AI Output Validation, Retry Policies
- Real-Time and Data Integration: Server-Sent Events (SSE), WebSocket, Streaming APIs, Streaming UI, Electronic Health Records (EHR), Fast Healthcare Interoperability Resources (FHIR), Epic, Cerner
- Frontend Architecture and Platforms: Modular Frontend Architecture, Reusable Component Systems, Workflow Automation, Workflow Orchestration, Workflow Canvas, Multi-Tenant B2B Platforms
- Enterprise Data and Security: Compute Node Management, Data Collaboration Systems, Data-Source Management, Role-Based Access Control (RBAC), Audit Logging, Multi-Party Computation (MPC)
- State Management: Vuex, Pinia, Redux, Zustand
- Testing and Quality: Jest, Cypress, Playwright, End-to-End (E2E) Testing, Browser-Based Test Automation
- Cloud and DevOps: Microsoft Azure, Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Docker Compose, Nginx, CI/CD

EDUCATION

Syracuse University

M.S.E. in Computer and Information Science, 2018

